

Machine Learning–Enhanced Fischer–Tropsch Synthesis: Optimizing Catalysts and Process Conditions for Efficient Fuel Production

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Fischer–Tropsch synthesis (FTS) offers a promising route for producing clean, renewable fuels. Yet, designing efficient catalysts and determining optimal process conditions remain major hurdles. Machine learning (ML) provides powerful means to address these challenges. Despite their potential, metal/zeolite catalysts are scarcely studied in ML-driven FTS research. This work applies an ML-based framework to model and optimize metal/zeolite catalysts for liquid fuel synthesis via FTS. Supervised learning methods reveal key structure–performance correlations, whereas multi-objective optimization identifies ideal catalyst and process parameters. The top solution is benchmarked against nearest experimental data. Results show CatBoost as the best-performing model, with Pt–Co/Beta treated with NaOH and NH_4^+ emerging as the optimal catalyst.

Keywords: Fischer–Tropsch synthesis, Liquid fuel, Machine learning, Zeolite

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1 Introduction

The increasing global demand for sustainable and cleaner energy sources has increased research into alternative fuel production pathways [1]. Among these, Fischer–Tropsch synthesis (FTS) is playing an important role in producing long-chain hydrocarbons from syngas (a mixture of CO and H_2 derived from biomass, coal, or natural gas) [2]. The performance of the FTS process is critically dependent on the nature of the catalyst, influencing not only the conversion efficiency of CO but also the distribution of hydrocarbon products [3, 4]. Metal/Zeolites are very attractive options due to their ability to modulate product selectivity toward fuel-range hydrocarbons (C_{5+}) while suppressing undesired methane formation. Zeolites play an important role in integrating FTS and hydrocracking as they hydrocrack waxes to liquid fuel hydrocarbon range products [5]. The unique acidity, shape-selectivity, and tunable pore structure of zeolites, when combined with active transition metals such as Co, Fe, or Ru, offer a versatile platform for catalyst design [6, 7]. However, the rational design and optimization of such bifunctional catalysts remain challenging due to the complex interplay between metal type, support properties, reaction conditions, and other physicochemical factors.

Traditional catalyst development approaches, based on trial-and-error experimentation, are time-consuming and resource-intensive [8]. In recent years, data-driven methods, particularly machine learning (ML), have emerged as

powerful tools to accelerate catalyst discovery by uncovering hidden structure–activity relationships, enabling high-throughput screening, and guiding the design of new formulations [9, 10].

The integration of ML with advanced optimization strategies is a rapidly progressing field, particularly relevant for process engineering and heterogeneous catalysis. ML-driven surrogate models combined with Bayesian optimization have been employed to efficiently explore vast compositional spaces and converge and greatly reduce computational cost [11]. Similarly, sampling of Pareto frontiers using regression and classification models has been shown to balance exploration and exploitation, thereby identifying Pareto-optimal designs under complex constraints [12, 13]. Multi-objective evolutionary algorithms such as NSGA-II offer a powerful framework for catalyst optimization, particularly when trade-offs between competing objectives—such as activity, selectivity, and stability—must be considered simultaneously [14]. This powerful technique has also been

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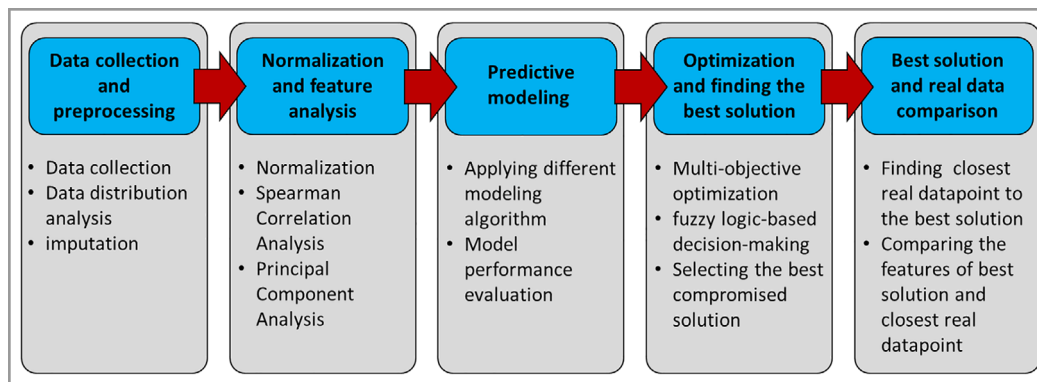


Figure 1. Modeling and optimization framework.

used for the FTS process. Chakkingal et al. [15] showed that artificial neural network regression is able to capture complex non-linearity in FTS kinetic data. They used SHapley Additive exPlanations (SHAP) techniques to find process variable ranking in model prediction. According to this study, temperature and pressure are the most important input variables for light olefin selectivity, and space–time velocity and syngas ratio have lower effect. Motaev et al. [16] used random forests as a predictive tool to predict FTS performance for cobalt catalysts. They reported surface area, pressure, and temperature as the most important features. Teimouri et al. [17] also applied random forests to predict CO conversion and C_{5+} selectivity for CO hydrogenation to clean liquid fuels with the focus on carbon supports. A very strong positive correlation is detected between the pore size of the catalyst, support material, and the catalyst performance. This is because support materials with larger pore sizes improve the dispersion of the active sites and positively affect the chain growth. This study also showed that parameters, such as promoter loading, reduction pressure, and H_2/CO ratio, have lower effect on catalytic activity and selectivity. Shafiee et al. [18] have focused on ML application in FTS with emphasis on C_8 – C_{16} hydrocarbon production, reporting that H_2/CO ratio, temperature, and surface area are key to jet fuel selectivity, and H_2/CO ratio, reduction, and reaction temperatures are key factors for conversion.

Despite this growing interest, the application of ML techniques in FTS catalyst optimization remains underexplored. Although metal/zeolite catalytic systems are very important for liquid fuel production purposes, they have not yet been modeled and optimized using ML. This work presents an ML-based framework for the predictive modeling and multi-objective optimization of metal/zeolite catalysts aimed at maximizing CO conversion and C_{5+} selectivity. The framework of modeling and optimization is described in Sect. 2. The results of modeling and promising catalyst composition and operating condition identification are then presented in Sect. 3. This approach not only enhances the understanding of the governing parameters

in FTS catalysis and provides a generalizable methodology for data-driven catalyst design in complex systems but also presents an optimum catalyst and operational condition for liquid fuel production through FTS.

2 Methodology

In this work, a metal/zeolite catalytic dataset is modeled and optimized. The goal is to maximize CO conversion and C_{5+} hydrocarbon selectivity using ML and optimization techniques. The proposed methodology includes the following steps: data preprocessing, exploratory analysis, predictive modeling, and multi-objective optimization. Python programming language is used for modeling and optimization, and the major steps are detailed below, as shown in Fig. 1.

2.1 Data Collection and Preprocessing

The dataset is collected from published experimental studies on liquid fuel production using metal/zeolite catalysts. It includes different parameters such as catalyst composition and structural and physicochemical properties, as well as the corresponding FTS performance metrics (CO conversion and C_{5+} selectivity). Distribution of each feature is studied, and the missing values are identified. The missing values are handled using data imputation techniques. Different imputation methods such as statistical methods (mean, median, or mode) and more advanced approaches (e.g., automated imputation), are assessed for this purpose. JMP software is used for this preprocessing analysis. Categorical variables (e.g., support, active phase metal, and synthesis method) are encoded using label encoding.

2.2 Normalization and Feature Analysis

To ensure consistent input scales for modeling, variables are normalized. To identify the descriptors that have higher

effect on the catalyst performance, Spearman correlation analysis is performed to examine monotonic relationships between descriptors and decision variables. In parallel, principal component analysis (PCA) is applied to reduce dimensionality and visualize the structure of the data, aiding in identifying clusters or trends within the dataset.

2.3 Predictive Modeling

Before modeling, the dataset is split into training (80 %) and testing (20 %) subsets to ensure generalizability and reproducibility. Different ML algorithms are implemented to develop predictive models for CO conversion and C_{5+} selectivity based on catalyst properties and operating conditions. CatBoost [19] and light gradient boosting machine (LightGBM) [20] are nonlinear, tree-based gradient boosting methods that model complex relationships by constructing an ensemble of decision trees. The number of estimated parameters is not limited to the feature count but instead is a function of the number of trees (often several hundred to thousands), their depth, and leaf weights, typically resulting in a large, data-dependent parameter set. Support vector regression (SVR) [21], although based on linear principles in its native form, most commonly uses a nonlinear kernel, with the effective number of parameters corresponding to the number of support vectors (often in hundreds), their coefficients, and selected hyperparameters like kernel width and regularization terms. This results in all three models capturing complex structure–property relationships, in contrast to classical linear regression, which estimates a fixed, relatively small set of coefficients. These models are evaluated based on performance metrics, including the coefficient of determination (R^2) and the root mean square error (RMSE). Cross-validation is used to ensure generalization.

2.4 Multi-Objective Optimization and Best Solution Selection

Multi-objective optimization is employed using the concept of the Pareto front to maximize CO conversion and C_{5+} selectivity. NSGA-II algorithm is used for this purpose [22]. As NSGA-II is a minimization algorithm by design, the negative values of the predicted objectives are minimized. This algorithm is implemented using the pymoo library in Python [23]. After convergence, the algorithm produces a set of Pareto-optimal solutions representing the trade-off frontier between decision variables.

Among the non-dominated solutions on the Pareto front, a fuzzy logic-based decision-making method is applied to identify the “best” compromised solution [24]. The fuzzy approach is applied to integrate subjective preferences (e.g., relative importance of CO conversion vs. C_{5+} selectivity) into a formal framework, enabling the selection of a single optimal point that best satisfies both criteria.

2.5 Comparing the Best Solution and Real Data Points

The catalyst formulation with the highest fuzzy optimum score is benchmarked against experimental data. To quantitatively identify the most similar experimental case, all real datapoints are mapped into a normalized feature space, and Euclidean distances are computed between each real point and the fuzzy-optimal solution. The real datapoint with the minimal distance is deemed the closest match, providing a rigorous, multi-feature basis for direct comparison.

This systematic methodology supports robust predictive modeling and optimization by ensuring feature-level comparability between predicted and real catalyst cases. The approach reliably translates model insights into practical catalyst screening and optimal condition selection.

3 Results and Discussion

3.1 Data Collection and Preprocessing

The dataset was collected from 96 experimental studies on liquid fuel production through FTS using metal/zeolite catalysts. Three different metals (Co, Fe, and Ru) and seven different zeolites (Beta, Y, ZSM-5, mordenite, SSZ-13, SSZ-22, and ZSM-11) are included. The dataset consists of 96 datapoints, with 26 descriptors on catalyst formulation and characteristics and 2 decision variables, which are CO conversion and C_{5+} selectivity.

Fig. 2 shows the missing data distribution of the dataset, with 7 % of the data missing. To handle these missing data, automated imputation in JMP software is used, as this approach is more accurate and has better distribution compared to the statistical methods.

3.2 Normalization and Feature Analysis

Data are normalized to provide homogeneous input scaling for modeling, and the importance and the effect of different descriptors on the decision variables are examined using the Spearman correlation analysis, which is depicted in Fig. 3.

According to this analysis, the active phase percentage and pore volume are the most effective parameters on CO conversion, and descriptors such as calcination parameters and promoter and support pretreatments have negligible effects. As for C_{5+} selectivity, ion exchange is positively affected, and zeolite SiO_2/Al_2O_3 ratio and the reduction temperature are among the parameters that are negatively affected. However, important descriptors, such as support, temperature, and pressure, are not recognized as highly effective parameters. This is because the Spearman correlation analysis does not show nonlinear or non-monotonic relationships, and it does not capture feature interactions.

To have a better understanding of the key parameters in the data structure and to reduce the dimensionality of the

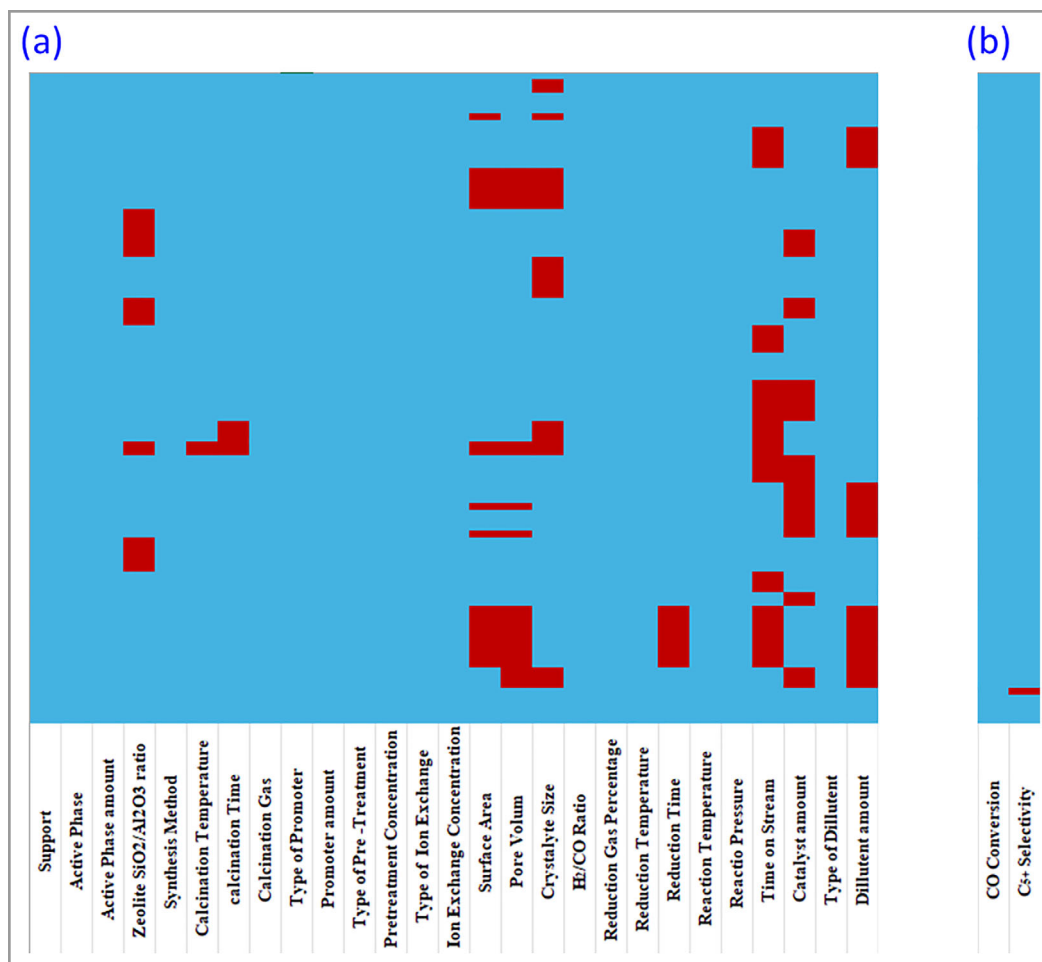


Figure 2. Missing data distribution of the FTS dataset for (a) descriptors and (b) decision variables.

dataset, PCA is applied. As shown in Fig. 4a, the first four principal components captured approximately 60 % of the total variance.

The PCA component loadings (Fig. 4b) reveal which original features contribute most significantly to each principal component. By examining the top contributing variables for each of the first principal components, insight is gained into which features drive the most variation in the dataset.

In PC1 and PC2, parameters, such as the active phase, surface area, zeolite $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio, reaction temperature, and pressure, have higher effect on CO conversion and C_{5+} selectivity. It means they positively affect liquid fuel production. Parameters, such as the synthesis method, calcination gas, and dilutant amount, have a low effect on decision variables.

The findings from PCA analyses guide the preparation of the dataset for predictive modeling. By excluding descriptors with negligible or redundant influence and retaining those identified as more influential, the feature space is refined for the modeling stage. This way, the ML models in Sect. 3.3 are

trained on a reduced but more meaningful set of variables, improving both interpretability and predictive performance.

3.3 Predictive Modeling

Tab. 1 presents the predictive performance of CatBoost, LightGBM, and SVR in modeling the two key catalytic outcomes: CO conversion and C_{5+} selectivity. Model training and validation are conducted using k -fold cross-validation. The R^2 and RMSE are reported as criteria for comparing the models.

Among these algorithms, CatBoost has the best performance with an R^2 of 0.99 (RMSE: 0.01) during training and 0.95 (RMSE: 0.05) during validation for CO conversion. Similarly, it attains an R^2 of 0.99 (RMSE: 0.02) in training and 0.90 (RMSE: 0.06) in validation for the C_{5+} selectivity. The relatively small performance drop between training and validation reflects the algorithm's strong generalization capability.

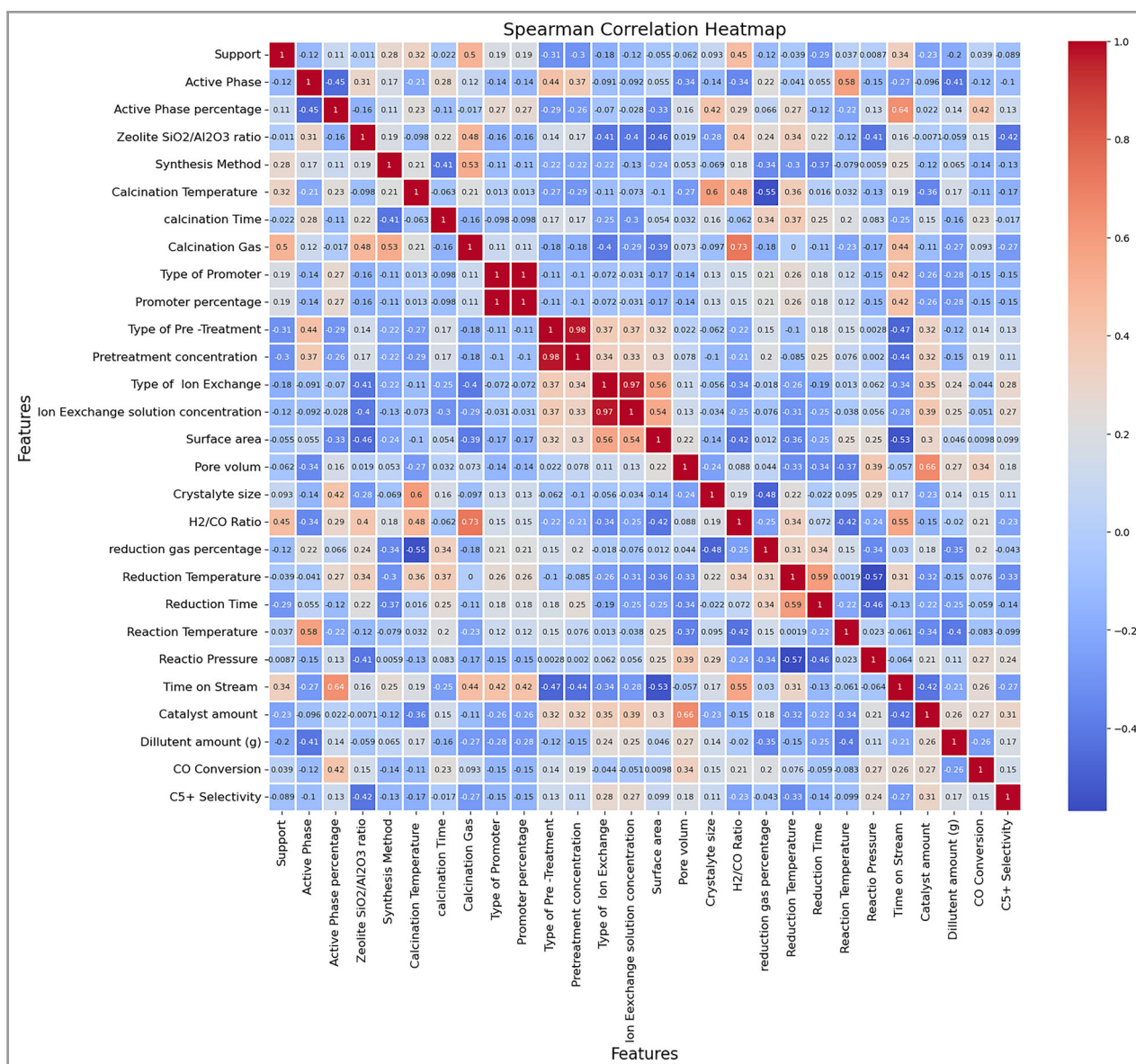


Figure 3. Spearman correlation analysis.

The performance of the LightGBM model is also promising, particularly for CO conversion (R^2 : 0.94, RMSE: 0.05 in validation), although the results for C₅+ selectivity are slightly worse (R^2 : 0.83, RMSE: 0.07). In comparison, SVR, which is a kernel-based regression model, underperforms in comparison with the other models investigated.

3.4 Multi-Objective Optimization and Best Solution Selection

After selecting CatBoost as the best predictive model, multi-objective optimization and fuzzy logic are applied to identify the optimal catalytic conditions for the FTS by maximizing CO conversion and C₅+ selectivity.

Fig. 5 shows the optimized Pareto front, which demonstrates the trade-off relationship between normalized CO conversion and C₅+ selectivity. The red points represent non-dominated (Pareto optimal) solutions, each constituting an optimal compromise: No point is objectively superior without incorporating preferences, and improvements in one decision variable typically reduce the other. The blue point represents the solution selected through the fuzzy logic-based decision-making approach, which integrates subjective preferences to identify a single best-compromised solution [24]. Thus, this point reflects a single preferred solution that aligns best with specified, providing a practical recommendation for catalytic process improvement.

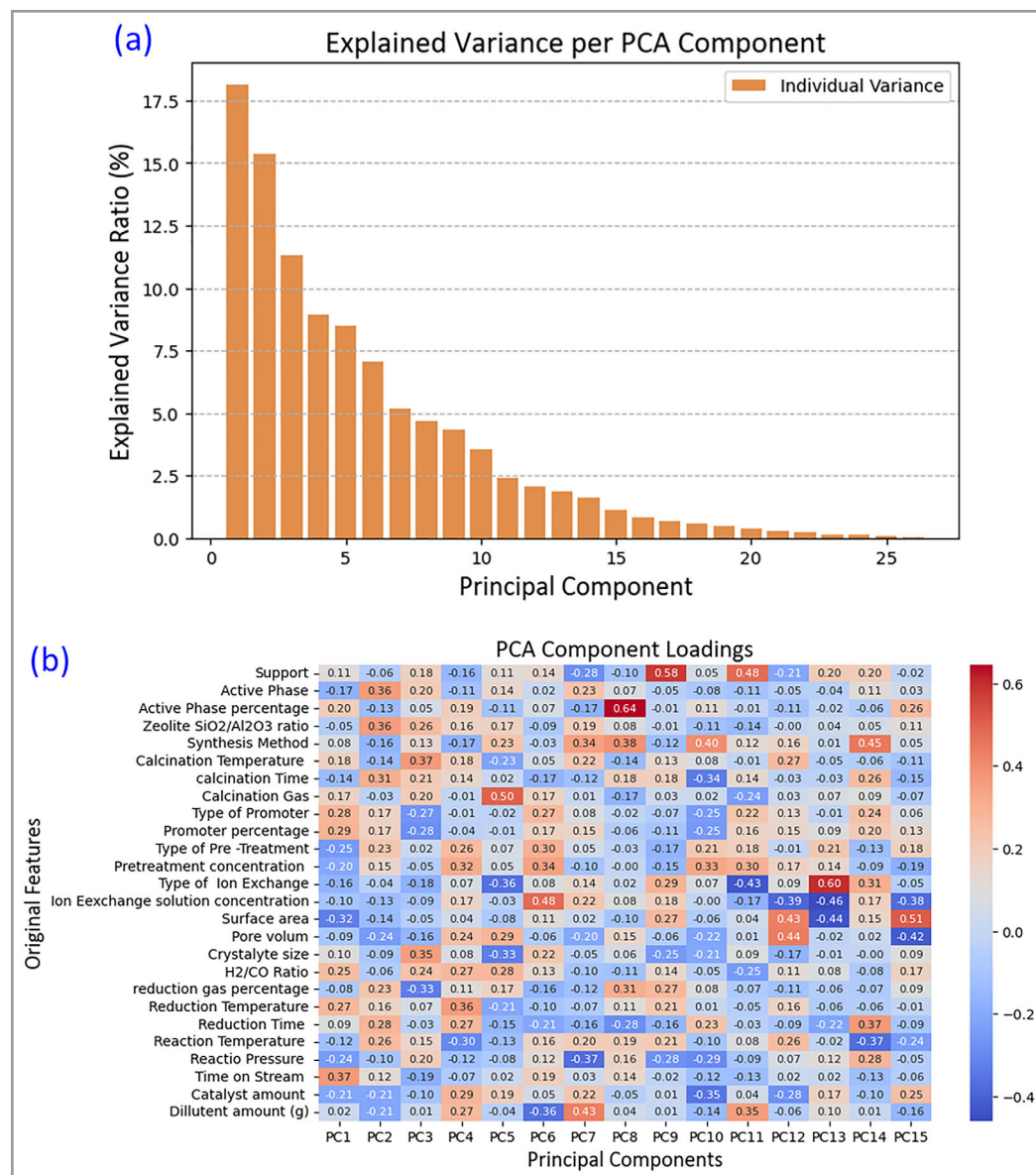


Figure 4. (a) Cumulative explained variance by PCA component; (b) PCA component loading.

Table 1. Predictive performance of evaluated predictive models.

Method	Decision variable	Number of parameters used in model	Training		Validation	
			R^2	RMSE	R^2	RMSE
CatBoost	CO conversion	9525	0.99	0.01	0.95	0.05
	C ₅₊ selectivity	9525	0.99	0.02	0.90	0.06
LightGBM	CO conversion	9500	0.97	0.04	0.94	0.05
	C ₅₊ selectivity	9500	0.89	0.06	0.83	0.07
SVR	CO conversion	2870	0.92	0.07	0.86	0.09
	C ₅₊ selectivity	2870	0.86	0.07	0.75	0.09

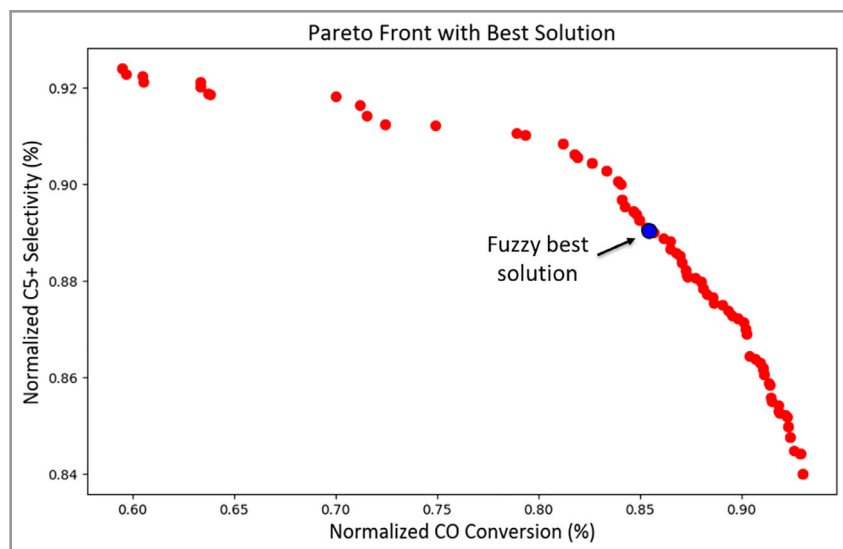


Figure 5. Optimized Pareto front with the best solution.

3.5 Comparing the Best Solution and Real Datapoints

Fig. 6 illustrates the comparison of the best fuzzy solution with real data, which are colored according to active phase and support. Following the procedure described in Sect. 2, the real datapoint exhibiting the smallest Euclidean distance to the optimized solution in the normalized feature space is selected as the closest. This method captures overall similarity across all relevant catalyst and process variables. CO conversion and C_{5+} selectivity for the best solution and closest real point are comparable, demonstrating the modeling approach's practical relevance.

A detailed feature comparison of both cases is summarized in Tab. 2, which highlights their differences and similarities, including catalyst formulation details and operating conditions. This analysis demonstrates that the optimization process produces solutions that are realistic and closely aligned with successful experimental outcomes.

The best optimized solution is a Pt-Co/Beta catalyst with 15 wt % Co as active phase and 0.03 wt % Pt as promoter. The zeolite support of the optimized catalyst also includes pretreatment with NaOH and ion exchange with NH_4^+ . According to the literature, such treatments improve mesoporosity and regulate the acidity,

which positively affect liquid fuel production [25]. Optimum reaction temperature and pressure are 240.35 °C and 19.66 bar.

The closest real datapoint is prepared by mixing powders of Co- Al_2O_3/SiO_2 catalyst (35 wt %), H-form of Beta zeolite (30 wt %), and boehmite (35 wt %) [26]. The active phase and zeolite support are selected to be the same as the best optimized solution. The active phase percentages are different, as is 15.08 wt % for the best solution and is 20 wt % for closest datapoint. The difference in zeolite SiO/Al_2O_3 ratio might be because of zeolite NaOH treatment in the optimized catalyst.

Reaction conditions, such as H_2/CO , reduction and reaction temperature, and reaction pressure of the best solution and

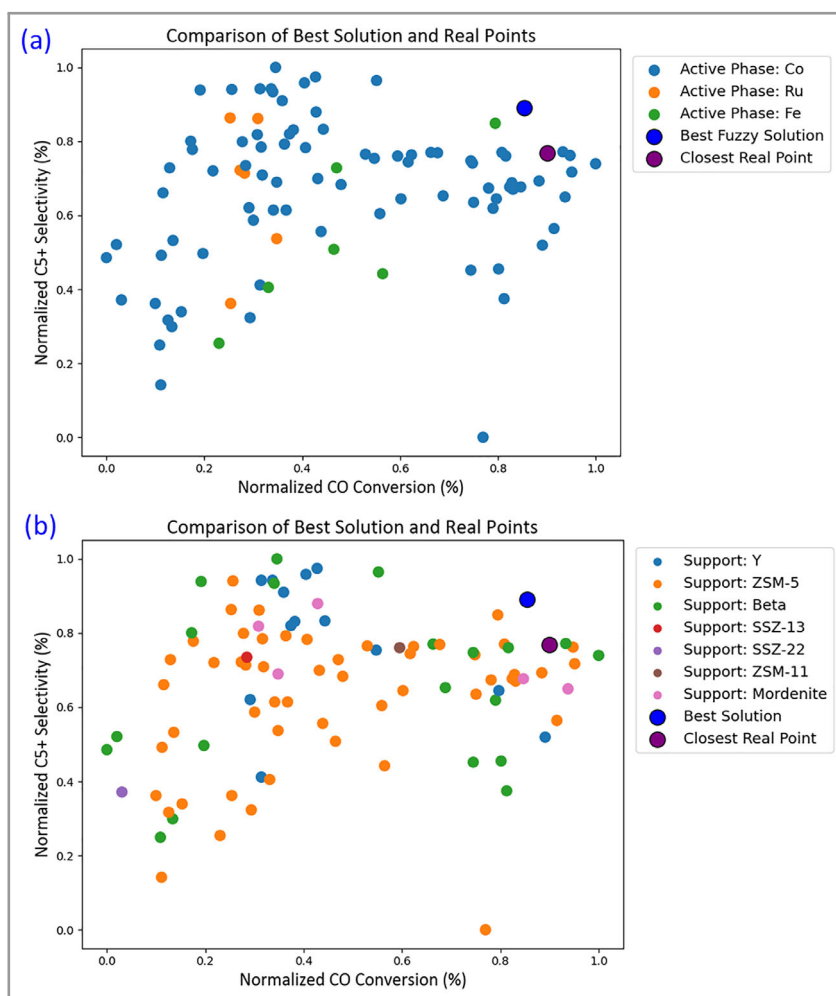


Figure 6. Comparison of the best solution and real datapoints that are classified with different colors according to the (a) active phase and (b) support.

Table 2. Feature comparison of the best solution and closest real datapoint.

Feature	Best solution	Closest real point
Support	Beta	Beta
Active phase	Co	Co
Active phase concentration [wt %]	15.08	20.00
Zeolite SiO/Al ₂ O ₃ ratio	20.86	40.50
Calcination temperature [°C]	366.69	400.00
Type of promoter	Pt	NA
Promoter concentration [wt %]	0.03	NA
Type of pretreatment	NaOH	NA
Pretreatment concentration [M]	0.05	NA
Type of ion exchange	NH ₄ ⁺	NA
Ion exchange concentration [M]	0.67	NA
Surface area [m ² g ⁻¹]	443.36	311.00
Pore volume [cm ³ g ⁻¹]	0.42	0.57
Crystallite size [nm]	20.01	14.70
H ₂ /CO ratio	2:1	2:1
Reduction temperature [°C]	398.42	400.00
Reaction temperature [°C]	240.35	240.00
Reaction pressure [bar]	19.66	20.00
Time on stream [h]	58.04	57.81
CO conversion [%]	88.62	81.70
C ₅₊ selectivity [%]	85.59	73.20

closest points are very close to each other. The values for these parameters are 2:1, 400 °C, 240 °C, and 20 bar, respectively. This shows that optimization is acceptable and within the sensible range of real data.

4 Conclusion

In this work, the application of ML methodology to model and optimize sustainable fuel production through FTS is assessed. Metal/Zeolite catalytic dataset is evaluated because it plays an important role in fuel-range hydrocarbon production via FTS, and it has not been studied previously.

Different ML algorithms are evaluated for their predictive accuracy. Among them, CatBoost shows the best performance and is selected and used for subsequent multi-objective optimization to simultaneously maximize CO conversion and C₅₊ hydrocarbon selectivity. To facilitate decision-making under multi-criteria trade-offs, a fuzzy inference system is developed to assess and rank the Pareto solutions, leading to the identification of a best-compromise solution. The best optimized solution catalyst is Pt–Co/Beta with zeolite support pretreatment and ion exchange. This fuzzy-optimal solution is then compared with the closest

real data. The operational conditions of the best solution and closest point are very similar.

The comparison between the model-derived optimal solution and the closest real datapoint not only provided practical validation of the optimization framework but also demonstrated the potential applicability of such computational approaches in guiding catalyst formulation and process conditions for enhanced FTS performance.

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Abbreviations

FTS	Fischer–Tropsch synthesis
LightGBM	Light gradient boosting machine
ML	Machine learning
PCA	Principal component analysis
RMSE	Root mean square error
SVR	Support vector regression

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